MDM2 Group Project 1, Group 4

Cell Arrangement and Thermal Optimisation in Lithium-Ion Battery Packs

Cameron Bidiuc, Akshat Wadhwa, Andrew Seymour, Austin Gong, Surkhab Lidhar

November 21, 2025

Abstract

Exceeding critical temperatures in lithium-ion battery packs can cause performance and safety issues in electric vehicles. This report investigates how different geometric arrangements of cells affect the maximum number of cells that can be safely packed without exceeding a critical temperature of 60°C. A two-node core–surface thermal model is used to simulate the heat generation and dissipation for three common packing configurations: Offset Columns (OC), Hexagonal Honeycomb, and Squares. The arrangements are evaluated under varying convective and conductive modes and inter-cell gaps to determine their thermal performance. Results indicate that when the parameters limit heat transfer in the lattice, geometry effects matter less since specific cell to cell heat transfer is constrained by weak boundary conditions, and the arrangements are comparable. However, for sufficient heat transfer conditions, the impact of specific geometries on overall performance is clearer, and OC performs best, showing geometric arrangement can have a significant increase energy density while maintaining safe operating temperatures, more so than the electrical configuration.

1 Introduction

Lithium-ion batteries have become the primary energy storage technology for electrical vehicles due to their high energy density and low self-discharge rate [1]. As the electric vehicle market expands, the demand for battery packs that are both compact and capable of delivering high power continues to grow. To meet these requirements, battery packs are assembled from multiple individual cells connected together in a combination of series and parallel arrangements.

Each cell generates heat through resistive heating, and if this heat is not dissipated effectively, the temperature can rise above critical levels. When the temperature of a cell exceeds 60°C the battery begins to degrade, leading to a loss of performance. In some extreme cases, it can also trigger the release of flammable gases which is a major safety hazard [2]. Single cells can easily dissipate heat to the surrounding air, but with multiple cells, the heating scales with pack size. This is because cells in a pack can pass heat to their neighbours, causing local hot-spots. Consequently, the geometric layout of cells has an important effect on heat distribution and what can be done to minimise overheating.

This report aims to address this challenge by trying to determine the maximum number of cells that can be safely used in a battery pack while remaining under the critical temperature threshold. The approach begins with modelling Joule heating in a single cell and then extends this to a multi-cell system. Using this model, multiple packing arrangements are tested and evaluated to identify which layouts support the highest cell counts. Finally this report investigates how physical design parameters, such as cell spacing, cooling coefficients and boundary heat loss affect different packing arrangements.

2 Modelling the Multi-Cell System

To understand how cells in the battery pack function, we need to model how a single cell works. In this section, we will build towards a multi-cell model to investigate the thermal behaviour of the battery pack.

The properties of the 18650 cylindrical lithium-ion cell were used to model the cells in this report because it is one of the most widely adopted and well-characterised cells in electric-vehicle and consumer-electronics applications. The designation "18650" refers to its dimensions: approximately 18 mm in diameter and 65 mm in height. This format has been used extensively in vehicles such as the Tesla Model S and Model X due to its reliable performance, and broad availability. Since each individual cell has a relatively small capacity, battery packs are typically assembled from many such cells connected in series and parallel. A standard 18650 cell operates at around 3.6 V with a typical capacity of roughly 3000 mAh.

2.1 Base Assumptions

The following are the base assumptions that apply throughout the entire project for the sake of simplicity:

- All cells are identical
- Thermal and electrical properties of cells remain constant
- Ambient temperature remains uniform
- There are no significant chemical reactions or phase changes in the cells
- Heat dissipation through radiation is negligible

2.2 Single Cell Model

In a battery, heat flow can be divided into internal heat generation and heat transfer. Most academic papers use the following equation to model the heat stored in a battery [3]:

$$\underbrace{C\,\frac{dT}{dt}}_{\text{Thermal Storage}} = \dot{Q}_{\text{gen}} - \dot{Q}_{\text{loss}} \tag{1}$$

where C is the equivalent heat capacity and Q is the rate of heat transferred.

There are 3 main types of heat conductivity: conduction, convection, and radiation. The effects of radiation can be negligible when compared to the heat transferred due to convection and conduction. From Newton's Law of Cooling [4]:

$$\dot{Q} = hA(T - T_\infty) \tag{2}$$

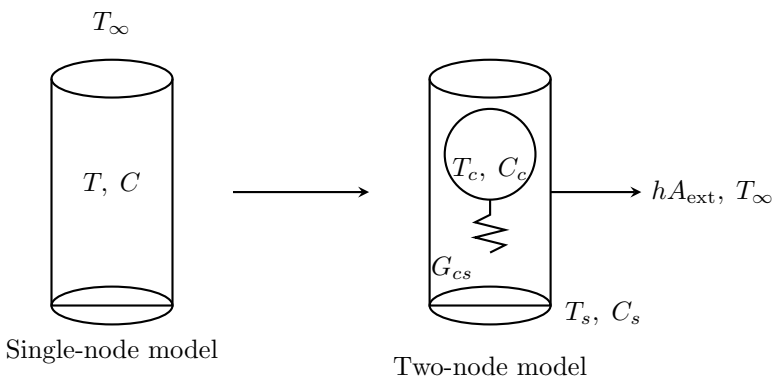
where T is the temperature, h is the heat transfer coefficient which roughly describes the cooling strength, A is the area over which the heat transfer is taking place and T_∞ is the temperature of the surrounding medium.

Therefore,

$$\underbrace{C\,\frac{dT}{dt}}_{\text{thermal storage}} = \underbrace{I^2R}_{\text{Joule Heating}} - \underbrace{hA\left(T - T_\infty\right)}_{\text{conduction + convection}} \tag{3}$$

This gives us a good first approximation of each cell's heating and cooling behaviour. However, it tends to under-predict the maximum temperature. To improve upon this we can consider the cell as consisting of 2 nodes, a core and a surface, each having a different temperature.

The Biot number is an important indicator for determining the number of nodes, whose value represents the ratio of internal thermal conductivity to external convective thermal resistance within the cell. The Biot Number for the 18650 cells indicates that internal temperature gradients are present, justifying the use of a two-node model. The details and justifications for this are provided in the Appendix.



This gives us coupled Ordinary Differential Equations (ODEs), one for the core and the other for the surface:

$$C_c \frac{dT_c}{dt} = \alpha \, \dot{Q}_{\text{gen}}(t) - G_{cs} \left(T_c - T_s\right), \tag{4}$$

$$C_s \frac{dT_s}{dt} = (1-\alpha) \, \dot{Q}_{\text{gen}}(t) + G_{cs} \left(T_c - T_s\right) - hA_{\text{ext}} \left(T_s - T_\infty\right). \tag{5}$$

where C_c, T_c refer to the capacitance and the temperature of the core, C_s, T_s for the surface. α refers to the fraction of the heat generated in the core, A_{ext} is the area of the surface of the cell and finally G_{cs} refers to the thermal conductance between the core and the surface.

2.3 Multi-cell model

To build up to the multi-cell model, we need to analyse each cell i by using the single cell model in addition to now considering any thermal conduction with its neighbouring cells. This is represented for a lumped 1-node cell by the following equation:

$$C_i \dot{T}_i = I_i^2 R_i - h_i A_i (T_i - T_\infty) - \underbrace{\sum_j G_{ij}(T_i - T_j)}_{\text{thermal conduction between cells}} \tag{6}$$

Where G_{ij} describes the thermal conduction between cells, given by,

$$G_{ij} = \frac{K_{ij} A_{c,ij}}{L_{ij}} \tag{7}$$

K_{ij}: thermal conductivity of the material between two cells.
$A_{c,ij}$: Contact distance between the two cells.
L_{ij}: Thermal conductivity distance.

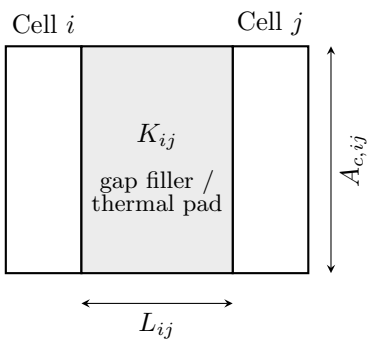


Figure 1: Geometric meaning of G_{ij} between two neighbouring cells

$$C_{c,i} \, \dot{T}_{c,i} = Q_i - k_{cs,i} \left(T_{c,i} - T_{s,i}\right) \tag{8}$$

$$C_{s,i}\,\dot{T}_{s,i} = k_{cs,i}\big(T_{c,i} - T_{s,i}\big) - g_i\big(T_{s,i} - T_\infty\big) - \sum_j G_{ij}\big(T_{s,i} - T_{s,j}\big). \tag{9}$$

Figure 2 illustrates the energy flow pattern under the 2-node model.

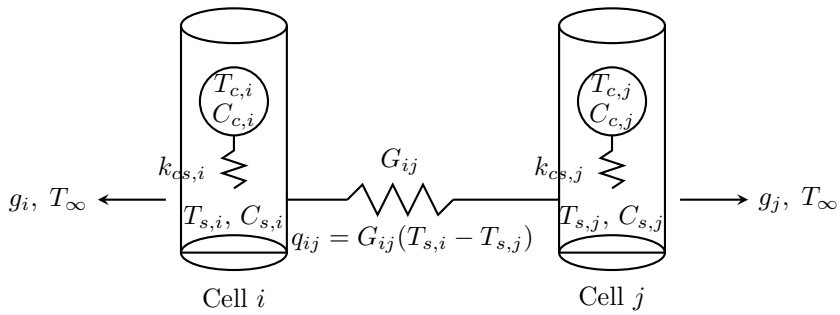


Figure 2: Multi-cell 2-node model highlighting thermal conduction G_{ij} between cell surfaces

3 Implementation of Cell Arrangements

We began by identifying a variety of potential arrangements based on repeating patterns, essentially making this part of our investigation a tessellation problem. These patterns were repeating squares, repeating hexagons, and offset columns. Offset columns is the cell arrangement used in industry [5], while hexagonal tiling is the most dense circle packing configuration [6], and repeating squares is the best at spreading air cooling through the battery module [7]. By generating these arrangements, we could model heat transfer and compare maximum cell results.

3.1 Square

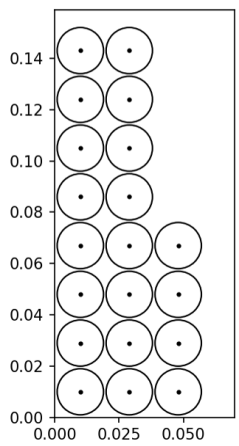


Figure 3: Square arrangement, $G = 0$ with column offset $D + G$

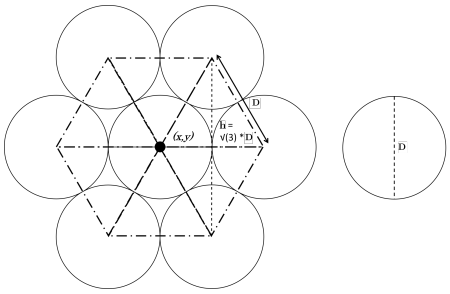


Figure 4: Hexagonal honeycomb base structure of height h, with cells of diameter D

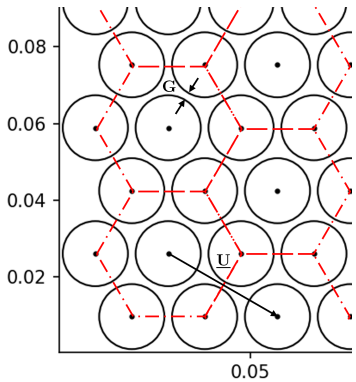


Figure 5: Hexagonal honeycomb arrangement, with cell gap G and column offset $\underline{U}$

The square arrangement is our base case; we arrange the cells using the most basic repeating model, which makes it easier for us to find the pattern. In addition, this arrangement also results in fewer cells per unit area, allowing for more space between the cells for cooling.

In our algorithm, we designed the size of individual cells and the spacing between cells, and then substituted them into the heat production and cooling equations to maximize the number of cells added while keeping the temperature below 60 °C.

3.2 Hexagonal honeycomb

The honeycomb packing is a hexagonal tiling of the module. The building block of this tiling is the hexagonal circle structure shown in Figure 4, which plots cells of diameter D at the center and each vertex of a regular hexagon of side length D and height $h = D\sqrt{3}$. In the actual tiling, however, we allow for variable spacing between cells for increased airflow.

Much like the offset columns, the honeycomb algorithm plots hexagons column by column, as illustrated in Figure 5. In each column, cells are plotted with a gap, G, between them in a stacked hexagonal structure until no more cells can be plotted without breaching the module boundary. The subsequent columns are offset by $\underline{U}$, and this process is repeated until the specified number of cells has been plotted.

$$\underline{U} = \begin{pmatrix} \frac{3}{2}(D+G) \\ -\sqrt{3}(D+G) \end{pmatrix} \tag{10}$$

3.3 Offset columns

The offset columns arrangement is the most optimal for module cooling as more of the cells' surface area is exposed and heat dissipates more evenly.

Our algorithm plots cells column by column, as illustrated in Figure 6. In each column, cells (with diameter D) are plotted with a gap, G, between them until no more cells can be plotted without breaching the module boundary. The subsequent columns are offset by $\underline{V}$, and this process is repeated until the specified number of cells has been plotted.

$$\underline{V} = \begin{pmatrix} D+G \\ (D+G)/2 \end{pmatrix} \tag{11}$$

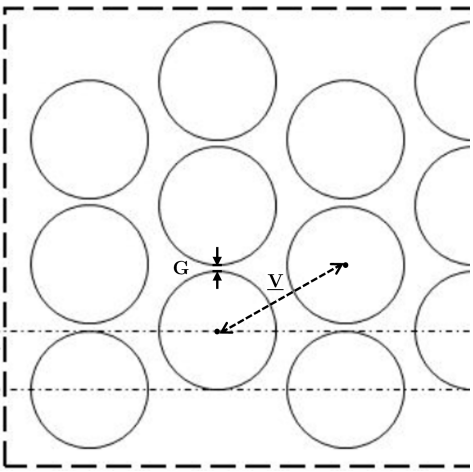


Figure 6: Offset Columns arrangement, with cell gap G and column offset $\underline{V}$

3.4 Implementation of the Laplacian and ODEs

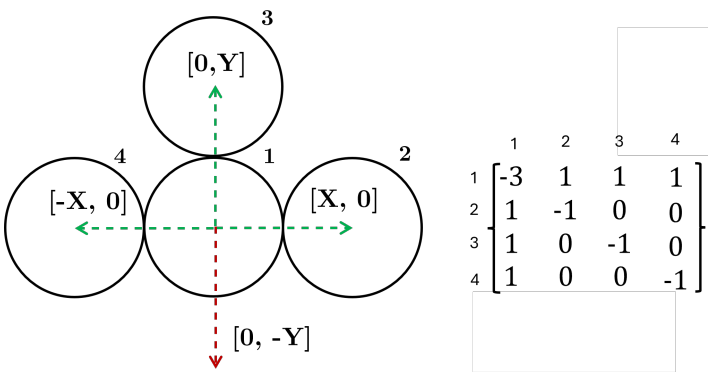


Figure 7: Laplacian example for a 4 cell square arrangement with neighbour detection vectors shown as green for detected, and red for not detected

The Laplacian was implemented by casting arrangement specific neighbour detection vectors out from a $cell_i$ to check for all possible neighbours. When detected the neighbour's index (calculated during the plotting of cells) is saved to a list for $cell_i$. This process is repeated for all plotted cells until the neighbours of each cell are known.

The laplacian matrix, L is then calculated by arranging these results into a matrix indexed by the cell indexes. By running a scan on python for each cell, if two cells, $celli$ and $cell_j$ are connected then $L_{ij} = L_{ji} = 1$. The diagonal entries are the number of neighbours that $cell_i$ has multiplied by negative 1.

An example of a simple laplacian for a 4 cell model is in Figure 7. The rows and columns sum to zero and this is a check employed in our calculations to validate results.

3.5 View Factor and Ray Tracing

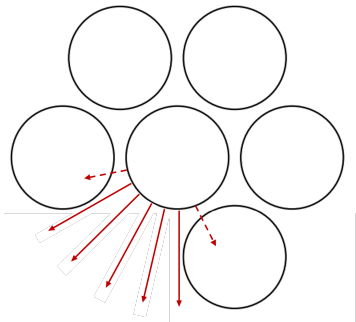


Figure 8: Heat emitted to the atmosphere by a honeycomb cell with 5 neighbours

For perimeter cells (i.e cells with less than the maximum possible number of neighbours), we had to model the heat dissipated to the boundary of the module. To do this, we used ray casting to determine the view factor, which is the *fraction of the radiation leaving surface i that is intercepted by surface j* [8]; here i is the cell surface and j is the module boundary.

By considering imaginary rays emitted normal to the surface of the cell in all directions (see Figure 8), we determine what proportion of rays aren't intercepted by the cells. Since each cell covers an equal portion of the cells emitted "rays", $1/max\ degree$. We then determine the general view factor of an arbitrary cell, c, to the module boundary, b, for any arrangement to be:

$$F_{cb} = 1 - \frac{cell\ degree}{max\ degree}. \tag{12}$$

Using this equation, we can determine the exposed surface area, S_A for any perimeter cell.

$$S_A = \pi D F_{cb} \tag{13}$$

Figure 8 shows that some of the rays may escape through the gaps between the cells, and this effect scales with G. However, when $G << 1$ there is little effect on the results for maximum cells since it only changes the view factor very slightly, and also some of the rays are absorbed by the lattice and don't escape to the module boundary, so under this condition eq. 13 is valid.

3.6 Electrical Configurations

Because the pack is required to deliver a fixed total power, the electrical configuration does not affect the total heat generated. Under the assumptions that all cells have identical and constant internal resistance, the total pack heat reduces to

$$P_{heat,total} = \frac{(P_{pack})^2 R}{n V_{cell}^2} \tag{14}$$

Every term in this equation is either constant or depends on the number of cells n, not on how those cells are arranged electrically. This is because series strings increase voltage and decrease current, while parallel do the opposite, but for a fixed total power output these effects cancel, leaving resistive heating unchanged. Therefore electrical configuration has no effect when total power output is fixed and the model should focus on the effects of geometric layouts instead.

4 Results

Running the model with differing physical (Gnn, h, Gcs, Cc, Cs, and T_∞) and environmental parameters ($T_{ambient}$) allowed us to visualize the performance of the arrangements over a wide range of scenarios, and better understand relationships between certain arrangements and parameters.

For the simulations, we had to define some base parameters. Figure 9 shows that during that for the base parameters, the performance of both the hexagonal honeycomb and offset columns is similar throughout, however as the temperature increases, the offset columns arrangement becomes the best one, allowing for a maximum of 64 cells against the 63 allowed for by the hexagonal arrangement. This is a demonstration of the effects of geometry limiting parameters as discussed later.

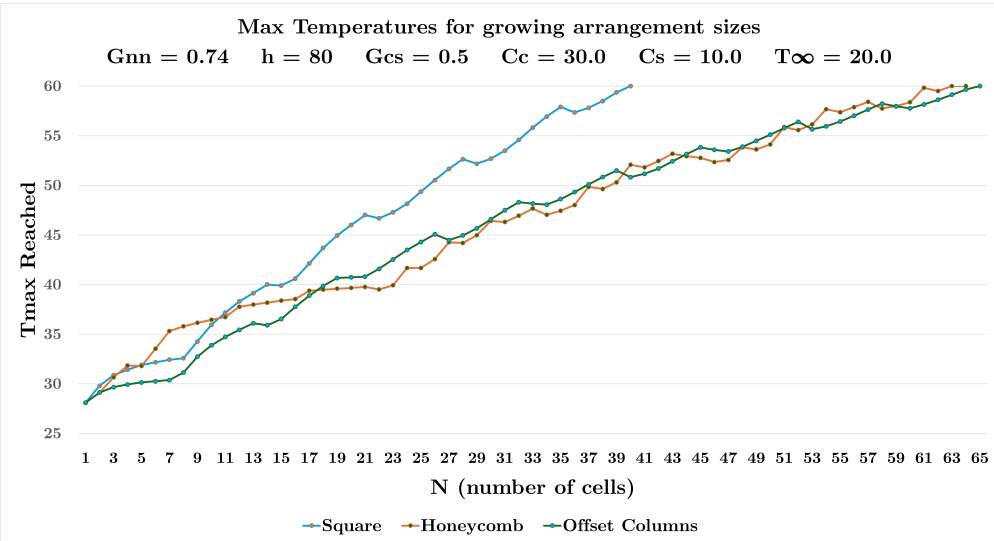


Figure 9: Max temperatures plotted against number of cells for all 3 arrangements using base parameters

4.1 Varying Parameters

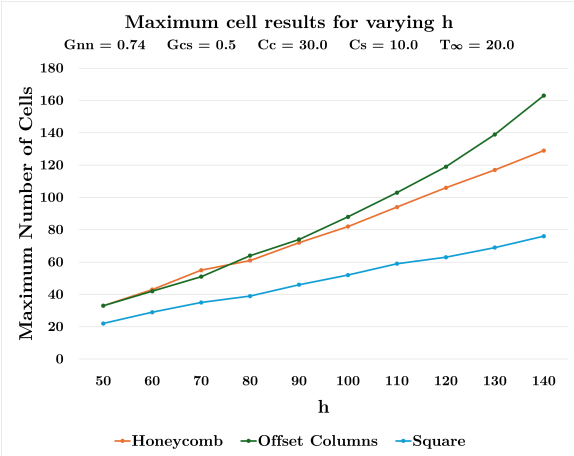


Figure 10: Varying h and the maximum cells output for each plotting arrangement

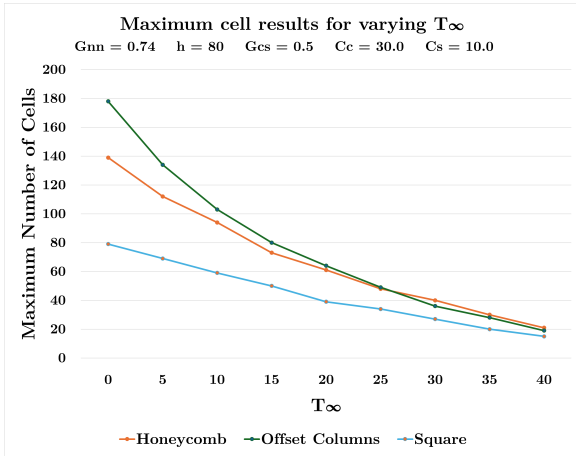


Figure 11: Varying $T\infty$ and the maximum cells output for each plotting arrangement

Figure 10 shows the performance of each arrangement as the cooling of external cells increases, where $h(\mathrm{W\,m^{-2}\,K^{-1}})$ is the convective heat transfer coefficient for perimeter cells to the surroundings. Simulating across $50 \leq h \leq 140$, our results show clear positive trends in the performance of each arrangement. The square and Honeycomb arrangements appear to follow a roughly linear correlation, while Offset Columns (OC) is more comparable to an exponential. This can be seen in the relatively higher performance of OC for larger h where while having comparable maximum cell output to the Honeycomb arrangement for $50 \leq h \leq 90$ (mean difference of 2), as h increases, OC becomes the superior arrangement by an average of 16.8 maximum cells for $100 \leq h \leq 140$.

Figure 11 demonstrates the impact of varying the environmental factors of our simulation on the results. We made $T_\infty = T_0 = T_{ambient}$ and then increased all values by 5 °C, to assess how different simulation temperatures might impact arrangement performance. Figure 11 shows almost mirrored results to Figure 10, however in Figure 11 Honeycomb and OC seem to respond more similarly to parameter changes, with Honeycomb behaving more exponentially here. This is reflected in the mean difference of 16 for $0 \leq T_\infty \leq 20$ compared to 16.8 for 10 last data points. Looking at the results as a whole, OC performs better than the other two arrangements for $0 \leq T_\infty \leq 25$ with an average 13.5 and 46.3 more cells produced than Honeycomb and square respectively, while for $30 \leq T_\infty \leq 40$ Honeycomb performs the best, with an average 2.7 and 9.7 more cells plotted than OC and Square respectively.

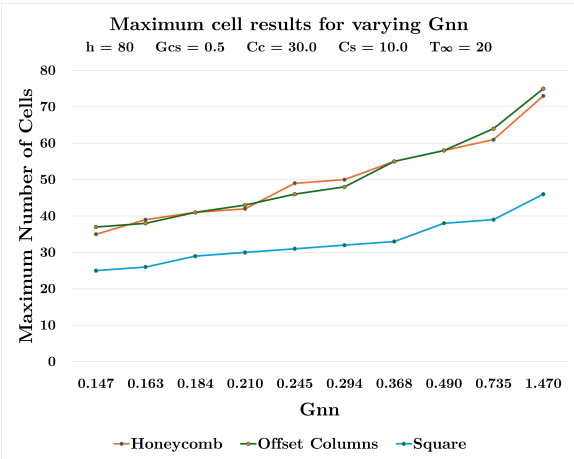


Figure 12: Varying Gnn corresponds to a 0.5mm increase of gap G starting at 0.5mm where h is fixed at 80

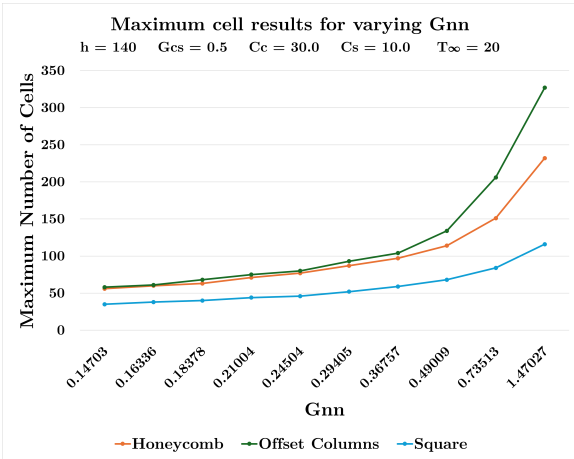


Figure 13: Varying Gnn proportional to a 0.5mm increase of gap G starting at 0.5mm where h is fixed at 140

Figures 12 and 13 demonstrate the impact of varying the gap, G, between cells where we simulated over $0.5mm \leq G \leq 5mm$ with a step of 0.5mm; the corresponding Gnn values, where $Gnn(\mathrm{W\,K^{-1}})$ is the neighbour-neighbour thermal conductance, calculated with eq. 7 and are shown on the x-axis of the figures. In Figure 12, $h = 80$, Honeycomb and OC perform better than Square, but perform very similarly to each other, where OC outputs an average 0.2 cells more than Honeycomb, suggesting OC is only very marginally more optimal under the specified parameters, with no discernible trend of increasing relative performance. However, in 13 we can see that while Honeycomb and OC still perform better than square for all Gnn, they perform differently to one another, where OC plots an average of 19.8 and 62.4 more cells than Honeycomb and Square, respectively. Of course, given the higher h the cells output across all arrangements is 1.97 times higher on average, and - adjusting for the scaling caused by favorable parameters - OC plots an average of 50 times more cells than Honeycomb and Square at $h = 140$ than at $h = 80$. Additionally, in contrast to Figure 12 we can identify an increasing trend in the performance of OC relative to the other arrangements as Gnn rises.

5 Discussion and Validation

5.1 Validation

Assessing the validity of our model, we considered the physical interpretations of our parameters which allowed us to better understand why our results are so. This allowed us to identify where our results don't agree with predictions and whether this meant we should question our model, or if it was an accurate reflection of reality.

5.1.1 Varying h

When looking at the results for varying h in Figure 10, the corresponding increase in performance of the arrangements is to be expected as by physically increasing the air cooling of the perimeter cells (what h represents) the diffusivity of temperature from the battery module would increase. The results also indicate that the Offset Columns arrangement also becomes increasingly better relative to the other models, this contrasts with empirical evidence [7] which suggests that more air cooling results in the square arrangement performing better, since heat is dispersed across cells more evenly. However, in our report air cooling isn't through the pack but rather just on the perimeter cells, so it is still reasonable to presume our model accurate, for varying h.

5.1.2 Varying temperature

Analysis of our results for varying the start temperature suggest that, for temperatures lower than 25, OC performs the best. This is to be expected since for lower temperatures the cell number can grow higher and geometry dependent cooling becomes more significant since cooling is more dependent on convection between cells; the OC arrangement has a thinner more, elongated arrangement than the other two and as a result has less deep interior cells making convection cooling more effective since heat reaches the perimeter easier. For temperatures above 25, Honeycomb performs marginally better, this is because as heat increases geometry dependent cooling decreases as the $hA(T_{cell} - T_{\infty})$ term in eq. 3 begins to diminish (convective heat loss becomes smaller). This term also explains the more general trend of convergence of the methods for increasing temperature, as the decreasing significance of the convective heat loss factor decreases the significance of specific geometry on results.

5.1.3 Varying neighbour conductance Gnn

The results for varying Gnn had more nuanced interpretations. Essentially by varying Gnn and h we are looking at the combined effects of varying internal heat spreading and varying the perimeter heat removal by the surrounding medium. For $h = 80$ the cooling effects on the boundary aren't that strong so the impact of effective heat dispersion (i.e. increasing Gnn) is limited by the h value, since the heat shared between cells has nowhere to diffuse to. The geometry, however, does become significant for higher h where the transfer of perimeter heat to the surroundings is fast due to stronger cooling - an increase in the $hA(T_{cell} - T_{\infty})$ term. In figure 13 the significant increase (multiple of 50) in performance of OC relative to the other methods from $h = 80$ to $h = 140$ is seen in it's tearaway from the other two arrangements, where OC is the most optimal arrangement as fewer deep interior cells allow for better heat conduction to the boundary.

5.1.4 Summary and Real-World Correlation

In summary, we see that OC performs better where simulation parameters geometry becomes more significant, and in general honeycomb and specifically OC perform significantly better than square and this is reflected in industry standard car batteries where staggered arrangements, similar to OC, are use such as in the Tesla model S battery modules where staggered columns are used but with varying gaps between sets of columns [5]. These findings further support the validity of our model and the accuracy of our conclusions.

5.2 Limitations

Several assumptions were made that simplified the model. The results provide reliable comparative trends but, they are not intended to fully match experimental battery pack behaviour.

Idealised Geometry: The model assumes perfectly packed, uniformly spaced cylinder, ignoring real structural elements like busbars, supports, and uneven gaps. This likely underestimates hotspots where heat does not spread efficiently.

Constant Thermal Properties: Thermal properties are treated as constant, even though they vary with temperature, pressure, and cell contact quality. This leads the model to underestimate sharp temperature rises in regions with poor contact or reduced cooling.

Two-Dimensional Heat Flow: The 2D nature of our model means heat flow is only simulated between neighbouring cells in the horizontal plane. This neglects axial conduction along the height of each cell which provides another path for heat to reach cooling structures.

Boundary-Only Cooling: Cooling is only applied at the perimeter, whereas real battery packs rely on dedicated cold plates that directly cool interior cells. This exaggerates the thermal advantage of elongated shapes and penalises compact geometries such as hexagonal packing.

5.3 Future Improvements

5.3.1 Add liquid cooling system

In our model, we used only air to cool the battery cells. While this is simpler in terms of material cost-effectiveness and model design, it is not the most effective cooling method. Adding a liquid cooled base plate would better represent a realistic car battery pack. We could have compared the efficacy of different coolants [9]. However, the cooling term would then dominate and substantially reduce the effect of geometry on thermal performance. Then the goal would be to determine the best spatial arrangement considering all the objectives when making a good battery pack.

5.3.2 Multi-node model

For a single battery cell, we used a 2-node model, which is a gray-box model between real-world conditions and purely theoretical calculations. Changing it to a multi-node model would more closely reflect reality. Furthermore, if liquid cooling is used, the temperature difference between the inside and surface of the cell will be very large, which would force us to use a multi-node model.

5.3.3 3D model

This study primarily investigates the arrangement and heat conduction of cells in a two-dimensional configuration, but reality is three-dimensional. If we incorporate longitudinal temperature analysis, three-dimensional arrangement, and more diverse cooling methods into

the 18650 battery cells, we would get results closer to experimental data.

6 Conclusion

By collecting results on maximum cells plotted by each arrangement over a range of parameters we were able to draw conclusions about the significance of geometric arrangements in determining maximum cell outputs, and most optimal arrangements. Our findings show that for situations where ambient temperature is low, or conduction between cells (Gnn) is high and perimeter cooling (h) is also significant enough for geometric configurations to matter, OC is the favourable arrangement. For high ambient temperatures, Honeycomb appears to be the better arrangement, but only marginally and this is due to reduced significance of the geometric configuration.

Overall, these results show that cell arrangement can meaningfully increase the maximum number of cells, but the optimal choice may depend on the thermal conditions. In thermally restricted scenarios, cell arrangement plays a smaller role, which demonstrates the importance of combining the geometric design with the right physical operating conditions to maximise engineering efficiency.

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[10] BatteryDesign.net. Series and parallel battery configurations, 2025.

A Example Hybrid Configuration Diagram

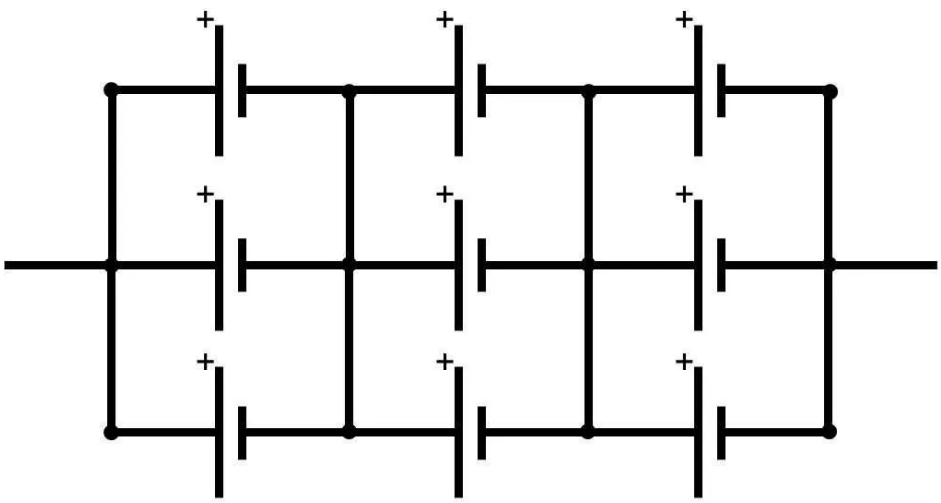

Figure 14: An example of a hybrid configuration. This one is a 3S3P meaning there are 3 cells connected in series in each branch, and 3 branches in parallel. Source: [10]

B Equations for electrical configuration derivation

$$n = n_s n_p. \tag{15}$$

$$I_{\text{cell}} = I_{\text{branch}} = \frac{I_{\text{pack}}}{n_p}. \tag{16}$$

$$V_{\text{pack}} = n_s V_{\text{cell}}. \tag{17}$$

$$P_{heat,cell} = (\frac{I_{\text{pack}}}{n_p})^2 R \tag{18}$$

$$P_{heat,total} = n_s n_p (\frac{I_{\text{pack}}}{n_p})^2 R \tag{19}$$

$$P_{heat,total} = \frac{n_s}{n_p} (I_{\text{pack}})^2 R \tag{20}$$

$$I_{pack} = \frac{P_{pack}}{V_{pack}}. \tag{21}$$

$$I_{pack} = \frac{P_{pack}}{n_s V_{cell}} \tag{22}$$

This equation onwards shows clearly how to get to our final equation

$$P_{heat,total} = \frac{n_s}{n_p}(\frac{P_{pack}}{n_s V_{cell}})^2 R \tag{23}$$

$$P_{heat,total} = \frac{(P_{pack})^2 R}{n_s n_p V_{cell}^2} \tag{24}$$

As $n = n_s n_p$, it leads to the final equation 14 shown in the report.

C Biot Number

The Biot number is an important indicator for determining the number of nodes, whose value represents the ratio of internal thermal conductivity to external convective thermal resistance within the cell.

$$Bi = \frac{hL_c}{k_{cell}} \tag{25}$$

where
h: Convective heat transfer coefficient
L_c: Characteristic length (in cells 18650, $L_c = \frac{V}{A} \approx 4 \times 10^{-3} m$)
k_{cell}: Equivalent thermal conductivity of the cell(It is usually between 0.5 and 1)

Then we chose to use air for cooling. The size of the battery under air-cooled conditions is approximately 100 $\mathrm{W \cdot m^{-2} \cdot K^{-1}}$. . [2]

$$B_i = \begin{cases} 0.8, & \text{when } k_{\text{cell}} = 0.5, \\ 0.4, & \text{when } k_{\text{cell}} = 1.0. \end{cases} \tag{26}$$

In cells, if the Bi number is greater than 1, a multi-node model should be chosen; if it is between 1 and 0.1, a 2-node model should be chosen; and if it is less than 0.1, a single-node model should be used. The Biot number is always between 0.4 and 0.8, so we chose 2-node model to analysis. [4]

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